

Agent-X

*A Proof Graph That Answers Questions
by Walking Its Own Structure*

Dimensions, Relations, and Projections

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Abstract

A proof graph $G \in \{0, 1\}^{N \times N \times R}$ where nodes are claims (nigamana) and typed edges are relations (visheshanam). The system derives answers to natural-language questions by walking its own structure — no external inference engine, no neural network at query time.

The graph declares: - An operation algebra $(\Sigma, \circ, {}^{-1})$ where each operation's eval name, arity, and inverse are graph edges (ganana, pratipaksha) — not external metadata - A mantra layer $\mathcal{M} = \{m_i : janya(m_i) \rightarrow phala(m_i)\}$ of typed functions expressed as krama s-expressions, spanning both physics and mathematics - A graded ring $R = \bigoplus_n R_n$ over sentence-grades with lexical morphism $\delta : \text{Words} \rightarrow V$ and intra-grade composition (avastha, visheshana, parimana) - A non-commutative visheshanam ring $\mathcal{V}$ with 10 generators governing edge composition - A bhasha (language) layer where morphology, verb forms, and word resolution are graph-native — universal concepts carry no language-specific edges

The philosophical ontology (Advaita Vedanta, Shaiva Tantra) is structurally grounded in the algebra: shunya = $0_{\oplus}$, pratipaksha = ${}^{-1}$, fixpoint iteration = pratibodha.

The Proof Graph

The proof graph is a 3-tensor:

$$G \in \{0, 1\}^{N \times N \times R}$$

Each node is a nigamana (truth-that-holds). The visheshanam ring $\mathcal{V}$ has 10 generators:

Generator	Symbol	Algebraic role
swarupa	$1_{\otimes}$	multiplicative identity
abheda	$\equiv$	equivalence
sthita	$\leq$	partial order
yukta	$+$	addition
siddha	$\vdash$	provability
kriya	$\times$	multiplication (non-commutative)
phala	$\rightarrow$	consequence
janya	$\leftarrow$	origin
drishthanta	$\exists$	witness
pratipaksha	${}^{-1}$	group inverse

Ring axioms: yukta = $\oplus$, kriya = $\otimes$ (non-commutative), shunya = $0_{\oplus}$, swarupa = $1_{\otimes}$, pratipaksha = ${}^{-1}$. The graph is self-describing: the node 'visheshanam' $\in G$ declares the ring structure using the same edge types it defines.

The Graded Ring and Lexical Morphism

A paragraph is a graded ring:

$$R = \bigoplus_{n=0}^k R_n$$

where R_n is the fact-set at sentence-grade n . Grade boundary = viraam (period). $\oplus$ is intra-sentence accumulation with four directions: vriddhi (add), kshaya (sub), multiplication, division. $\otimes$ is cross-sentence entity selection. Fold identity = $0 = \text{shunya}$.

The lexical morphism $\delta : \text{Words} \rightarrow V$ maps surface words to graph concepts:

$$\delta(\text{"heavier"}) = \text{viveka-max}, \quad \delta(\text{"flew"}) = \text{kshaya}, \quad \delta(\text{"find"}) = \text{vidhi-kaala}$$

δ is structure-preserving: it maps kshaya $\leftrightarrow$ vriddhi and preserves pratipaksha. Direction detection walks sthita edges on resolved verb nodes — the verb's graph structure determines the arithmetic operation, not a hardcoded table.

Intra-grade composition. Within a grade, adjacent words compose when the graph declares it. Three patterns:

1. **Avastha** (qualifier + concept): "initial" + "velocity" → initial-velocity. The concept's avastha edge lists valid qualifiers; composition fires when the compound node exists in the graph. 2. **Visheshana** (adjective + entity): "elastic" + "energy" → binds property to entity via visheshana edge. 3. **Parimana** (quantifier + entity): "each" + "bag" → distribution marker triggering $\otimes$.

Composition happens during construction (BQG reduce), not as a post-pass. The graph drives which words compose — no lookup tables.

The Operation Algebra

The operation algebra Σ is the set of all graph nodes carrying ganana (computation) edges. Each operation $\sigma \in \Sigma$ is declared entirely within the graph:

$$\sigma = (\text{ganana}(\sigma), \text{parse-arity}(\sigma), \text{pratipaksha}(\sigma))$$

where ganana maps to a primitive function, parse-arity gives argument count, and pratipaksha is the algebraic inverse — all as typed edges on the same node.

Pratipaksha is an involution on the symmetric core:

$$\text{pratipaksha}(\text{pratipaksha}(\sigma)) = \sigma$$

Verified pairs: add $\leftrightarrow$ sub, mul $\leftrightarrow$ div, square $\leftrightarrow$ sqrt, half $\leftrightarrow$ double, log $\leftrightarrow$ pow.

The pratipaksha relation is a graph edge — the inverse declaration lives in the same structure as the concepts it inverts. No external table; the graph IS the algebra. Resolution is $O(1)$ via a contract cache built at load time from graph walks.

Mantra Signatures

A mantra is a node m with typed function signature:

$$m : \text{janya}(m) \rightarrow \text{phala}(m)$$

The computation is expressed as a krama — a compositional s-expression over Σ :

$$K = \frac{1}{2}mv^2 \quad \Longleftrightarrow \quad (\text{krama half (multiplication mass (square velocity))})$$

$$T = \frac{2\pi}{\omega} \quad \Longleftrightarrow \quad (\text{krama division (multiplication 2 pi) angular-velocity})$$

$$\% = \frac{p}{w} \times 100 \quad \Longleftrightarrow \quad (\text{krama multiplication (division part-value whole-value) 100})$$

Mantras span two varga: physics-mantra (kinematics, energy, momentum, forces, waves) and math-mantra (percentage, with more planned). The varga walk is uniform — mantra-select finds applicable formulas regardless of domain.

Evaluation is recursive descent: tokenize the krama, resolve each op via the contract cache, apply with bound values. Chain derivation computes the least fixpoint:

$$\text{derive-chain} = \text{lfp}(\text{derive-step})$$

At each step, every mantra whose janya are all bound fires and binds its phala. Terminates by Kleene's theorem: monotone operator on a finite lattice.

Formula Inversion via Krama + Pratipaksha

Given mantra m with $\text{krama}(m) = \sigma_1(\sigma_2(\dots, x_i), x_j)$ and known phala c , solve for x_k :

Step 1. Extract the op-path from root to x_k :

$$\text{krama-path}(m, x_k) = [\sigma_1, \sigma_2, \dots, \sigma_n]$$

Step 2. Reverse:

$$[\sigma_n, \dots, \sigma_2, \sigma_1]$$

Step 3. Replace each σ_i with $\text{pratipaksha}(\sigma_i)$ and apply:

$$x_k = \sigma_n^{-1}(\dots \sigma_2^{-1}(\sigma_1^{-1}(c, b_1), b_2) \dots)$$

where b_i are the other arguments at each binary node, resolved from bindings.

Example. Solve $K = \frac{1}{2}mv^2$ for v :

$$[\text{half}, \text{mul}, \text{square}] \xrightarrow{\text{reverse}} [\text{square}, \text{mul}, \text{half}] \xrightarrow{\text{pratipaksha}} [\text{sqrt}, \text{div}, \text{double}]$$

$$v = \sqrt{\frac{2K}{m}}$$

Inversion is a **graph walk**, not symbolic algebra. The formula structure (krama) is data in G . The inverse operations (pratipaksha) are edges in G . A tantra reads both and composes them.

Logical inversion follows the same mechanism: $\wedge \leftrightarrow \vee$ via De Morgan, $\Rightarrow \mapsto \mapsto$ contrapositive. The walk is uniform across algebraic and logical domains.

Pipeline as Stratified Evaluation

The answer function is a composition of monotone strata:

$$\text{answer} = \text{emit} \circ \text{dispatch} \circ \text{detect} \circ \text{expand} \circ \text{refine}^* \circ \text{assert} \circ \text{construct}$$

Structurally equivalent to Datalog stratified evaluation. Each stratum S_i is monotone: $S_i(T) \supseteq T$ (only adds triples, never removes). The refine stratum is a krama-ordered chain of 13 sub-passes wrapped in fixpoint.

Within a grade, facts form a set (commutative, idempotent). Across grades, results compose relationally. Rules are graph walks, not logic clauses.

The Signal Bus

Three signal layers form an encoding/decoding channel:

$$\text{detect}(\text{encode}) \xrightarrow{\text{channel}} \text{dispatch}(\text{decode})$$

1. **Lexical** — $\delta(\text{word}) \rightarrow \text{node} \rightarrow \text{edges} \rightarrow \text{typed triples (satya, sankhya, mithya, copula, vidhi-kaala)}$ 2. **Grammar** — verb morphology emits kaala (tense) + vachana (number) on subjects 3. **Intent** — pattern detection writes signal triples into the question graph

The write-signal/read-signal contract: detect encodes intent, dispatch decodes it. In information-theoretic terms: encoding = detect, decoding = dispatch, channel = question graph, noise = mithya, redundancy = monotonicity.

S-Expression Syntax

Three file formats unified under s-expressions:

- **om5**: $\backslash(\text{layer name (relation target...) ...})$ — declares G . Nodes carry all their edges: swarupa, sthita, yukta, kriya, phala, janya, pratipaksha, ganana, naama, avastha, etc. - **tantra4**: $\backslash(\text{tantra name (params) body...})$ — computation as composition of named operations from Σ - **shabda**: legacy metadata format, being phased out — edges that were shabda key-value pairs are migrating to om5 graph edges (ganana, naama, sthita, etc.)

A tantra is a composition of named operations from Σ . Last expression is return value. Bindings are $\backslash(\text{name expr})$. With sufficient named helpers, tantra bodies read as sentences — compositions of words in the operation algebra.

Four abstraction layers: primitives $\rightarrow$ operations $\rightarrow$ compositions $\rightarrow$ pipeline. Each layer composes the one below.

Panchaavayava — The Proof Structure

Every answer follows the five-limbed syllogism (panchaavayava), the Indian proof structure from Nyaya:

$$\text{pratijna} \rightarrow \text{hetu} \rightarrow \text{udaharana} \rightarrow \text{upanaya} \rightarrow \text{nigamana}$$

1. **Pratijna** (thesis) — states given values
2. **Hetu** (reason) — states what is sought
3. **Udaharana** (example) — shows the mantra formula
4. **Upanaya** (application) — substitutes values into the formula
5. **Nigamana** (conclusion) — states the result

The structure arises from the graph walk. The phala chain $\text{hetu} \xrightarrow{\text{phala}} \text{udaharana} \xrightarrow{\text{phala}} \text{upanaya} \xrightarrow{\text{phala}} \text{nigamana}$ exists as edges in G .

Ontological Grounding

The philosophy IS the algebra — not metaphor layered on it. Three grounding levels:

Load-bearing (same node, two descriptions): shunya IS the additive identity $0_{\oplus}$ — the fold seed reads it. pratipaksha IS the algebraic inverse $^{-1}$ — the count-chain walks it. Delete these nodes and the pipeline breaks.

Level-indexed (one concept, multiple roles): purna $\equiv$ shunya at the $\oplus$ level AND purna $\equiv$ eka at the $\otimes$ level. Abhedha is level-indexed — the graph enforces this by omitting the shunya $\leftrightarrow$ eka edge.

Interpretive (structural analogy, no direct edge): pratibodha $\sim$ fixpoint, spanda $\sim$ eval. Philosophically compelling but computationally dispensable.

Results and Capabilities

The system answers questions across multiple domains by composing a small set of mechanisms:

Derive (mantra evaluation): forward computation via krama, inverse solving via pratipaksha walk. Physics mantras (kinetic energy, momentum, force, gravitational PE, wave speed, etc.) and math mantras (percentage). Chain derivation reaches fixpoint over multiple dependent formulas.

Count (grade fold): arithmetic word problems. The fold accumulates across grades with direction-sensitive operations (vridhhi $\rightarrow$ add, kshaya $\rightarrow$ sub, multiplication, division). Percentage scaling via shatamana. Distributivity handles "each" problems.

Compare (viveka projection): entity-scoped accumulation with argmax/argmin selection. Superlative and comparative forms resolve via graph structure.

Logic (boolean inference): categorical reasoning via swarupa/sthita walks, negation via pratishedha, "is X a Y" membership queries.

Unit composition: derive-unit builds dimensional algebra via fixpoint over krama chains. Every physics answer carries SI units derived from the graph — not hardcoded.

Morphology: verb forms generated at boot from dhatu seed nodes (kriya-nirmana). Plurals generated from vachana-nirmana. Nominalizations via naama edges. All morphology is graph-native.

Language independence: bhasha layer bridges surface language to universal sangati/kosha concepts. The same proof graph can serve multiple languages by swapping the bhasha layer.

Every answer is a panchaavayava proof. All computation is exact — no floating-point arithmetic.

Glossary

Graph Structure

Term	Meaning	Algebraic role
nigamana	truth-that-holds	node in G
visheshanam	relation type	edge label, ring generator
satya	truth-score	convergence measure via fixpoint
kosha	domain layer	typed subgraph
sangati	root layer	foundational concepts
bhasha	language layer	bridges surface words to universal concepts

Visheshanam (Edge Types / Ring Generators)

Term	Meaning	Symbol
swarupa	identity / IS	$1_{\otimes}$
abheda	non-difference	$\equiv$
sthita	foundation / rests-on	$\leq$
yukta	connection	$+$
siddha	proof / verified	$\vdash$
kriya	function / acts-as	$\times$
phala	consequence / produces	$\rightarrow$
janya	origin / born-from	$\leftarrow$
drishthanta	evidence / witness	$\exists$
pratipaksha	inverse / opposite	-1

Computation

Term	Meaning	Role
mantra	formula node	$m : \text{janya} \rightarrow \text{phala}$
krama	ordered sequence	s-expression over Σ
tantra	declarative program	composition of named ops
ganana	computation edge	declares eval name on operation nodes
naama	name edge	surface form of a concept
avastha	state/qualifier	valid qualifiers for compound composition
dhatu	verb root	seed node for morphological generation

Algebra

Term	Meaning	Role
shunya	zero / void	$0_{\oplus}$, fold identity
eka	one / unit	$1_{\otimes}$ at mul level
purna	fullness	$\equiv$ shunya at $\oplus$, $\equiv$ eka at $\otimes$
pratipaksha	inverse	involution: $\sigma^{-1^{-1}} = \sigma$
kshaya	decrease	$\Rightarrow$ sub in count fold
vridhhi	increase	$\Rightarrow$ add in count fold
viveka	discrimination	comparison operator
viraam	period / stop	grade boundary in R
shatamana	per hundred	percentage scaling in count fold

Pipeline / Inference

Term	Meaning	Role
anuvada	translation	reasoning layer
avrti	spiral return	fixpoint iteration
sandhi	joining	morphological composition
anumana	inference	categorical reasoning via varga
varga	category / class	type hierarchy for inference
mithya	false / noise	rejected triples
vidhi-kaala	imperative tense	seek/find signal
pratishedha	negation	negated concept edge
vibhakti	case relation	adhikarana, apadana, sampradana, karana

Proof Structure (Panchaavayava)

Term	Meaning	Step
pratijna	thesis	states givens
hetu	reason	states sought
udaharana	example	shows formula
upanaya	application	substitutes values
nigamana	conclusion	states result

Philosophy (Grounding)

Term	Meaning	Algebraic ground
shunya	emptiness	additive identity $0_{\oplus}$
pratipaksha	opposition	algebraic inverse $^{-1}$
pratibodha	awakening-to-awakening	fixpoint
spanda	self-pulsation	eval / computation
iccha	will / directed-reaching	directed graph walk

*The ring is the paragraph. The projection is the question.
Every answer is: build $\rightarrow$ fold $\rightarrow$ read.*